

Solidworks to URDF Export Guide

Ensure the Solidworks-to-URDF Add-In is installed

1. Download and install the sw_urdf_exporter from this link:

https://wiki.ros.org/sw_urdf_exporter

The last ROS 1 release Noetic will go end of life on May 31st with that the ROS Wiki (this website) will also be EOL and transition to being an archive. Maintainers: Please migrate any wiki content into your package's README.md file. If you need more help on migrating code please see this migration guide. Or watch Shane's Lightning Talk from ROSCon 2024.

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sw_urdf_exporter

SolidWorks to URDF Exporter

The SolidWorks to URDF exporter is a SolidWorks add-in that allows for the convenient export of SW Parts and Assemblies into a URDF file. The exporter will create a ROS-like package that contains a directory for meshes, textures and robots (urdf files). For single SolidWorks parts, the part exporter will pull the material properties and create a single link in the URDF. For assemblies, the exporter will build the links and create a tree based on the SW assembly hierarchy. The exporter can automatically determine the proper joint type, joint transforms, and axes.

[Download Installer](#)

If the above provided download fails to work on your system, please install by source before reporting an issue. If installing by source does work, please submit an [Update Installer Request](#)

ROS 2 Documentation

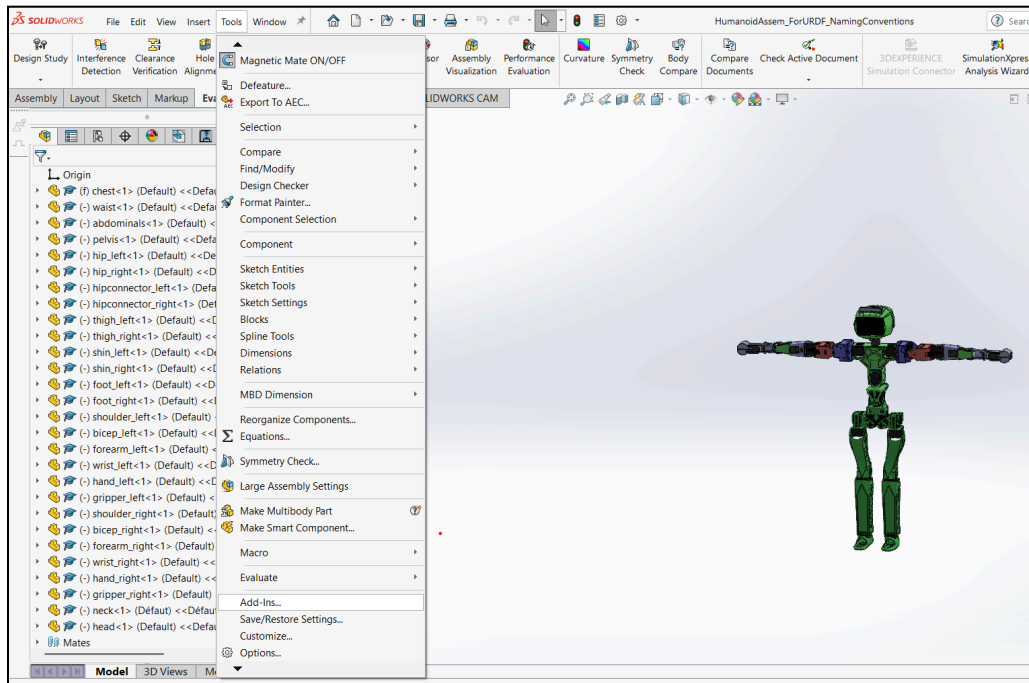
The ROS Wiki is for ROS 1. Are you using ROS 2 (Humble, Iron, or Rolling)?
[Check out the ROS 2 Project Documentation](#)
Package specific documentation can be found on index.ros.org

Wiki

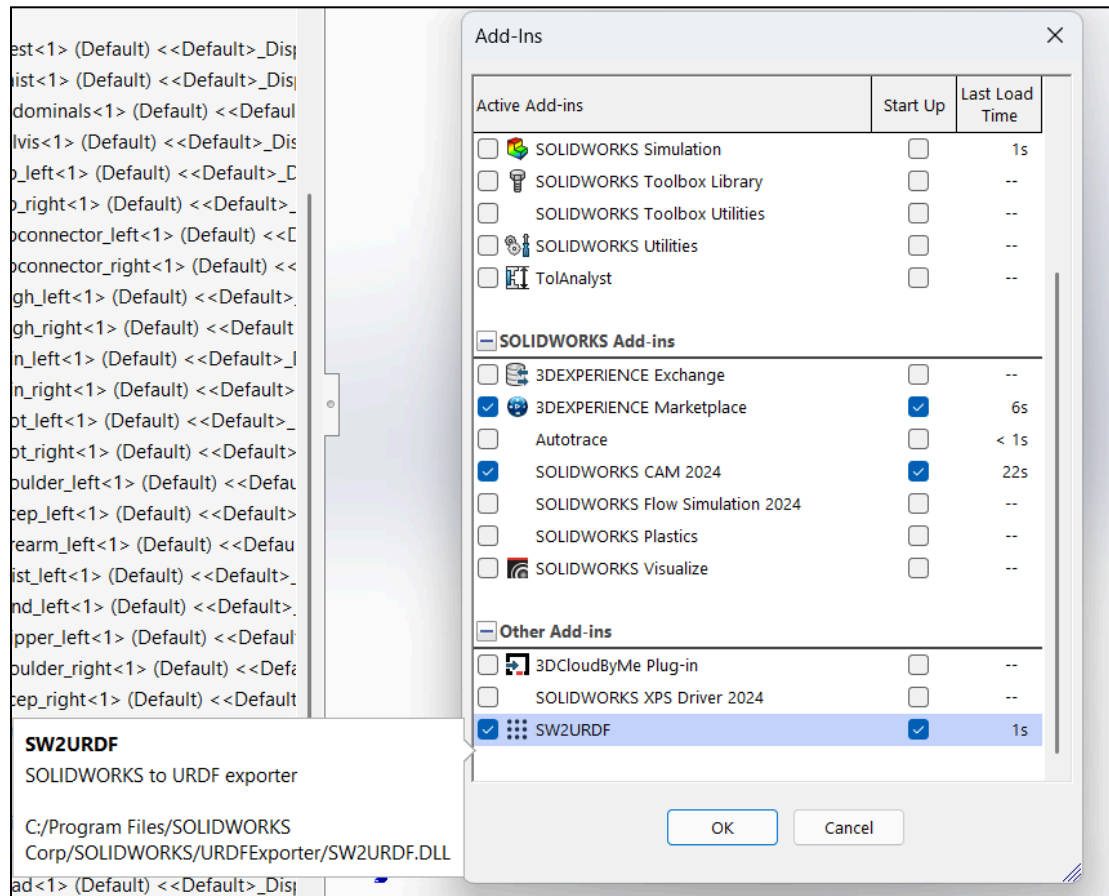
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2. Once you've downloaded and installed, open Solidworks and navigate Tools → Add-Ins



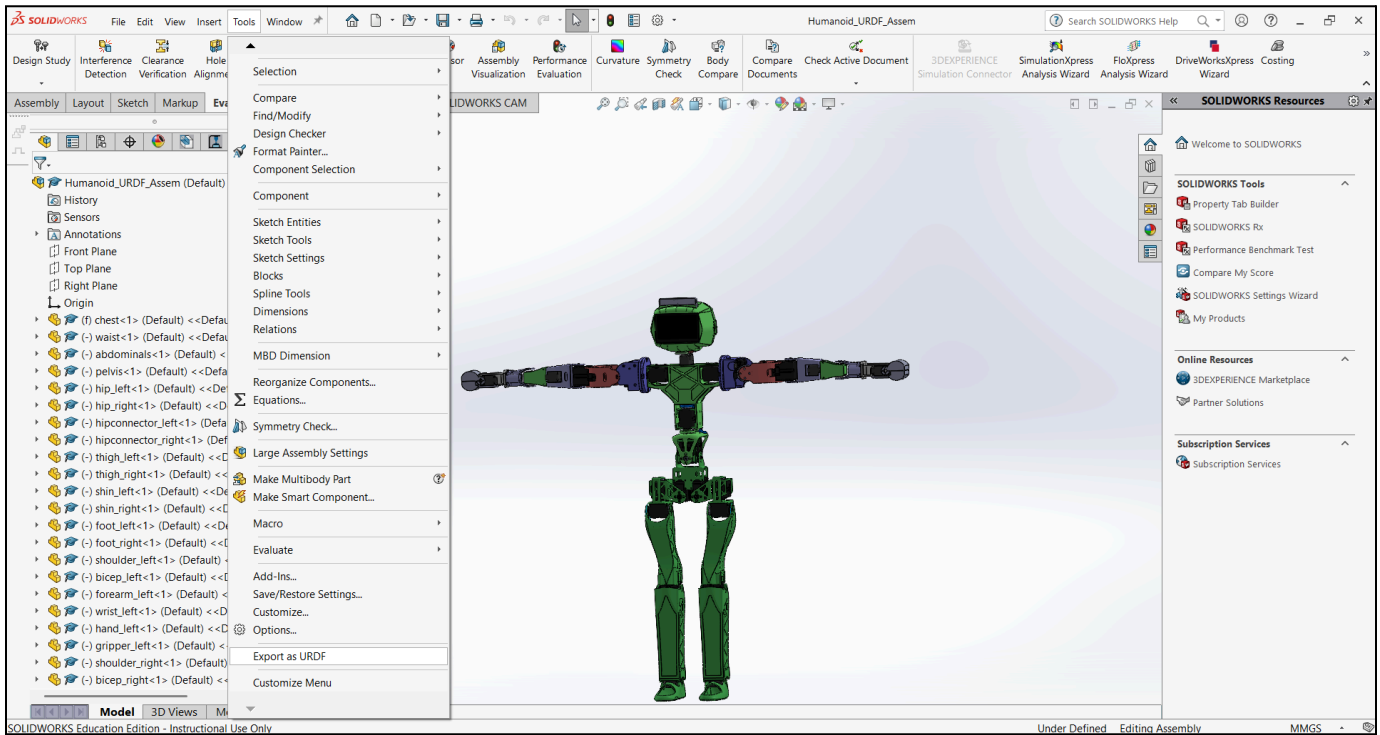
3. Activate the SW2URDF Add-In and click OK



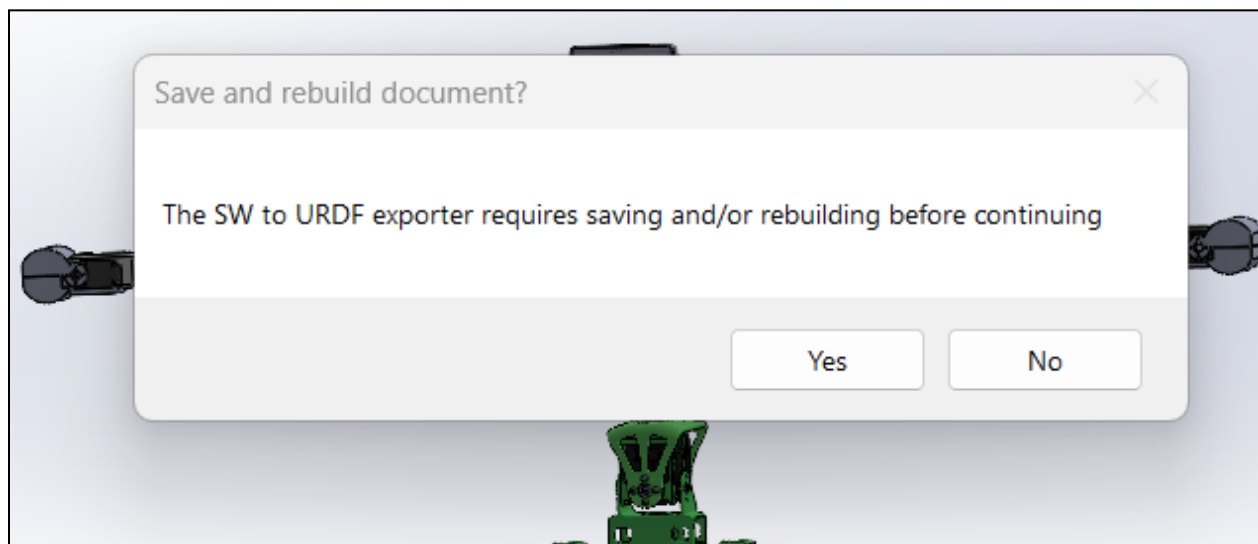
Export Solidworks to URDF

1. Navigate to Tools → Export as URDF

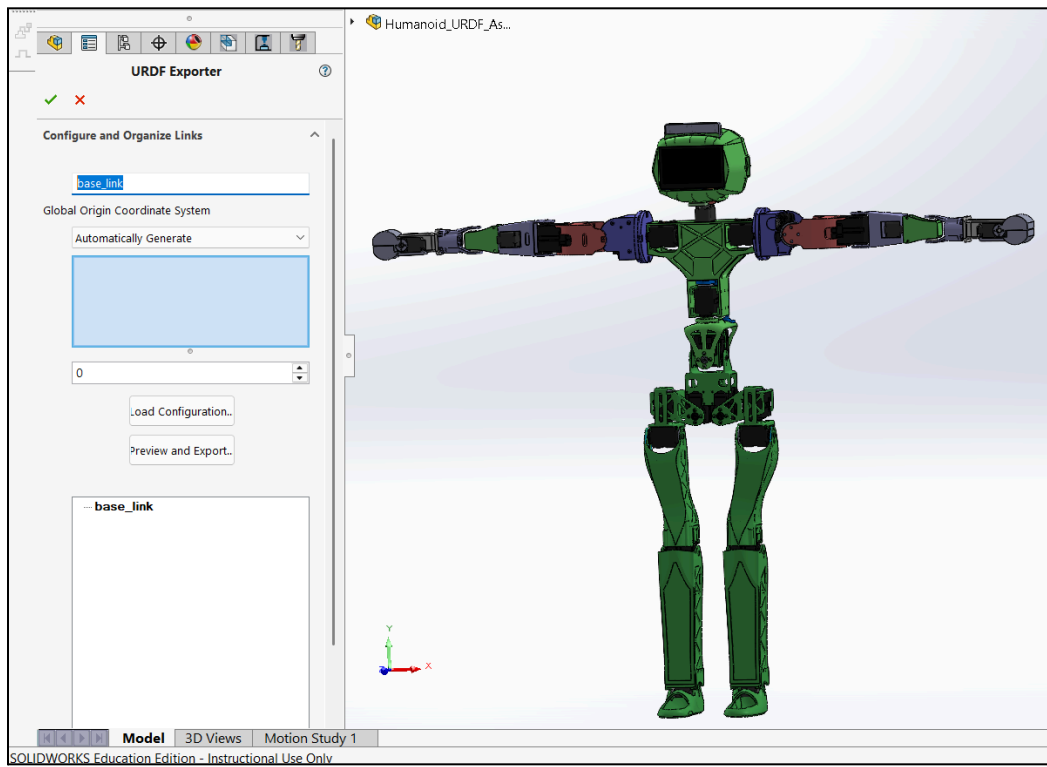
- You may have to hit the down arrow to find this Add-In after it is activated per the last section



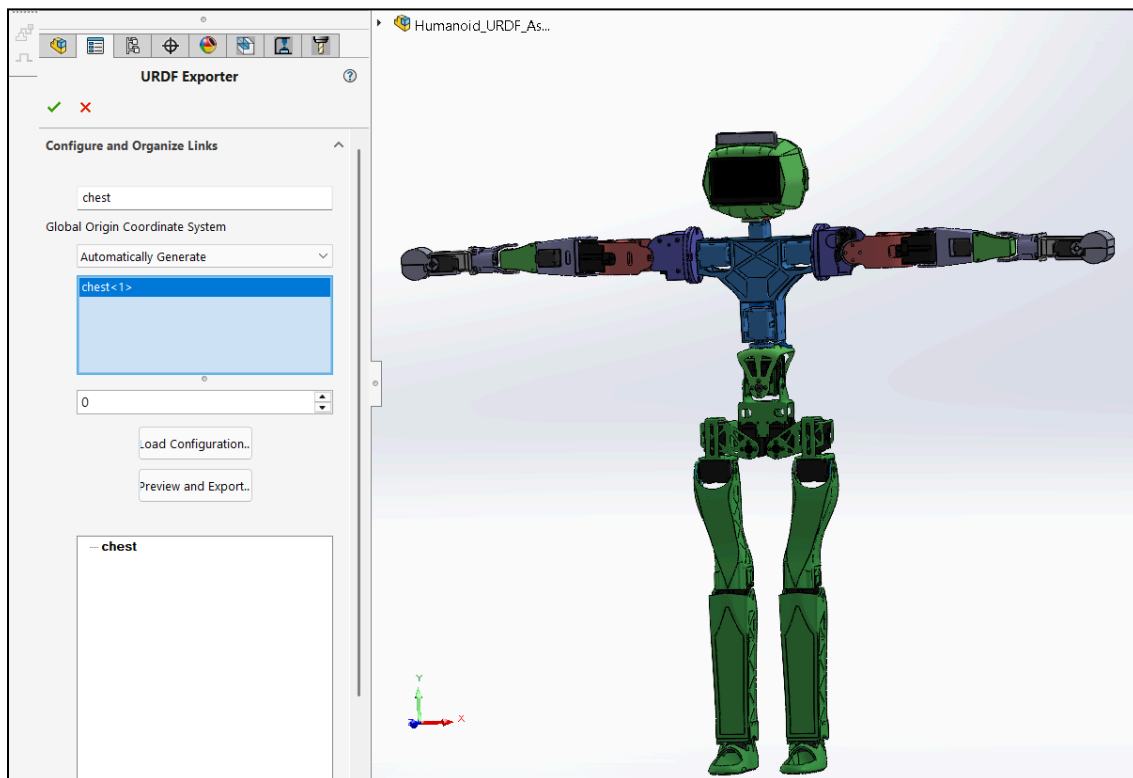
2. Rebuild the document if prompted



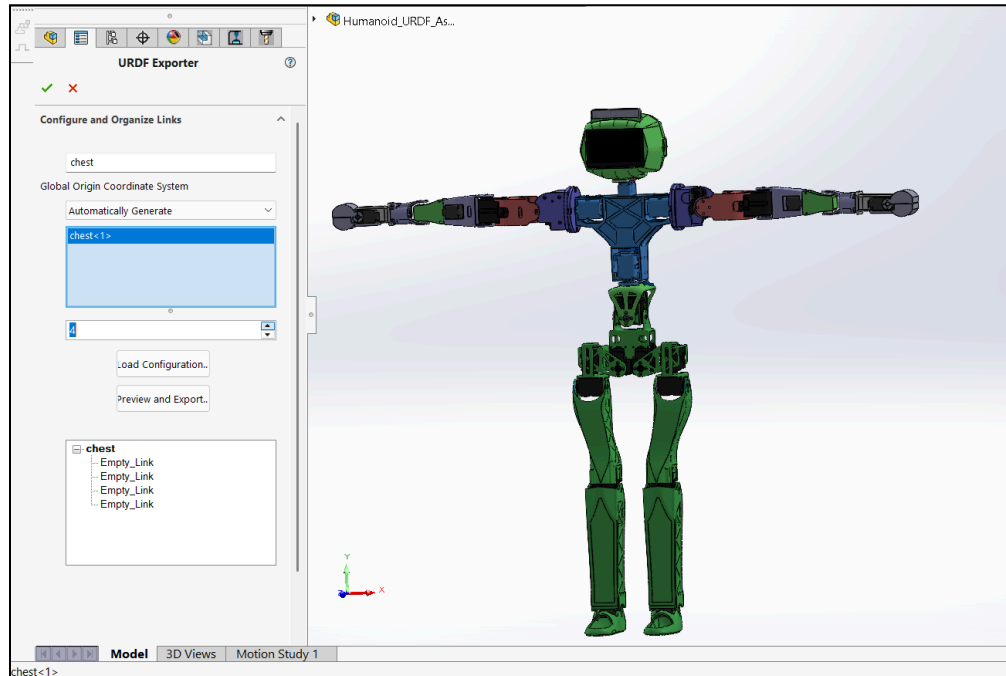
3. Rename “base_link” to “chest” in keeping with naming conventions



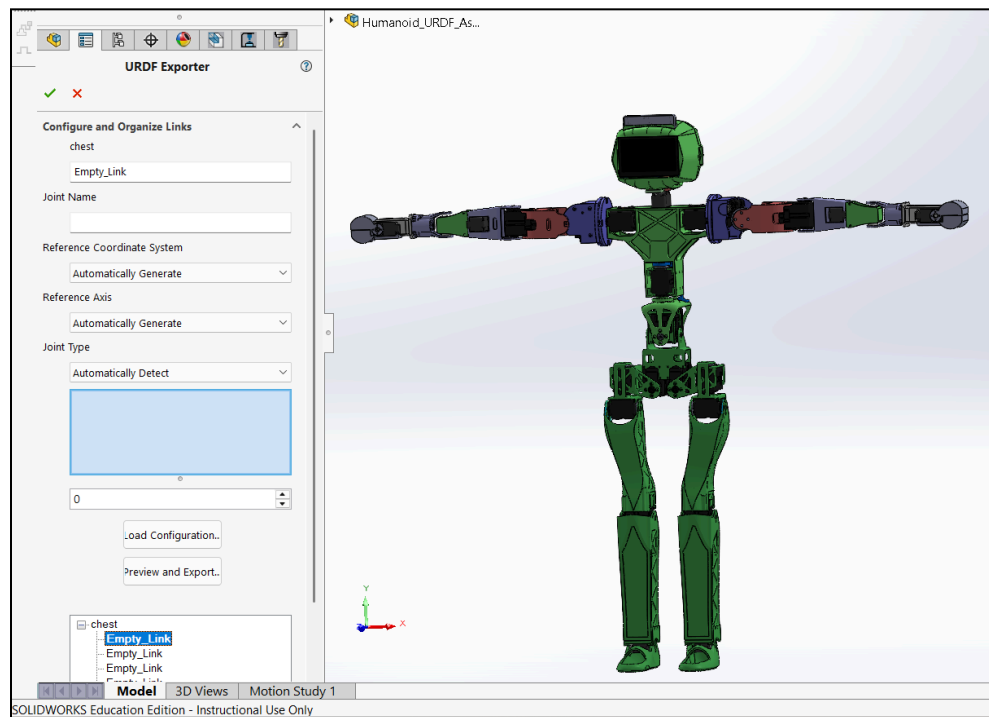
4. Select the “chest” link



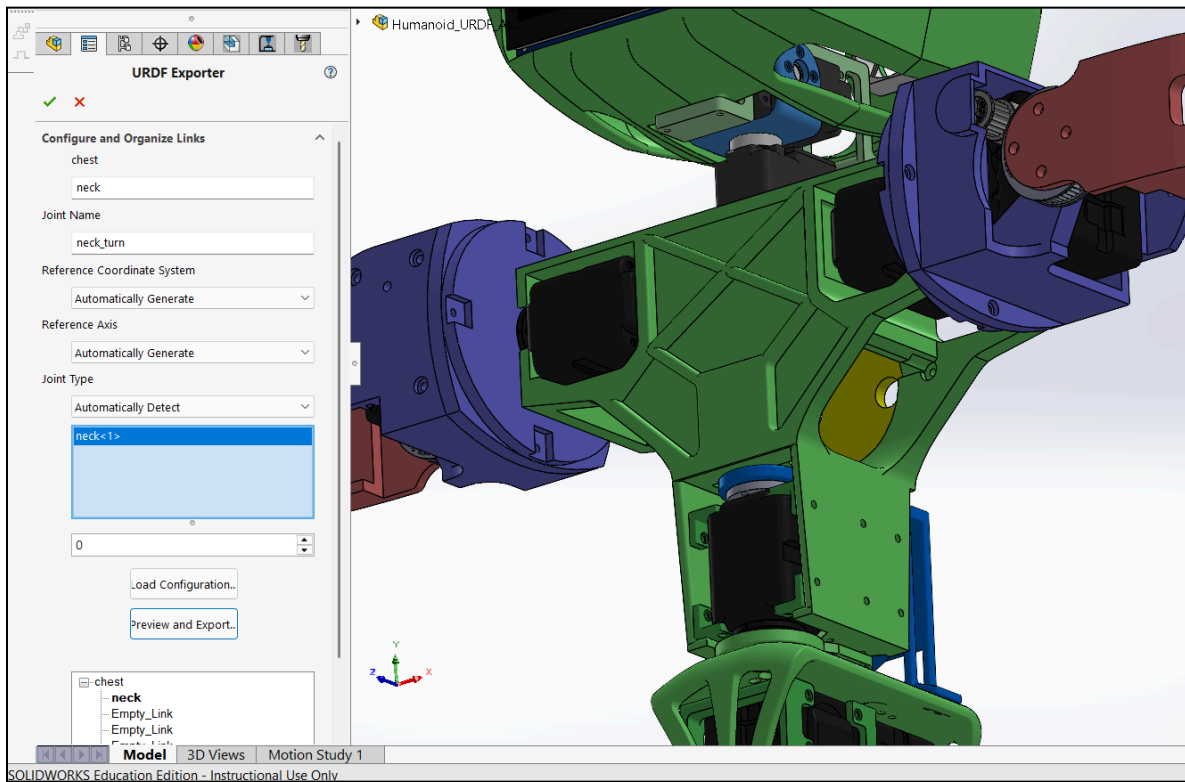
5. Add child links to the URDF tree based on how many joints branch from the base link. For this robot, there are 4 branches off of the chest (base link). These are the head, the right arm, the left arm, and the lower body.



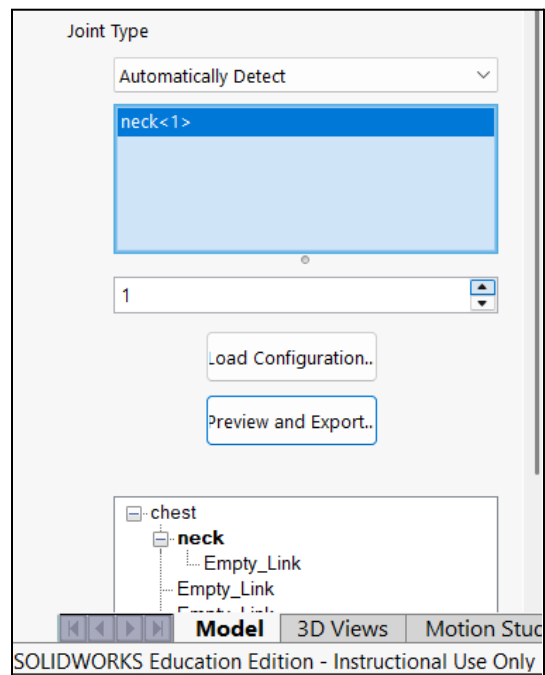
6. Select the first child link



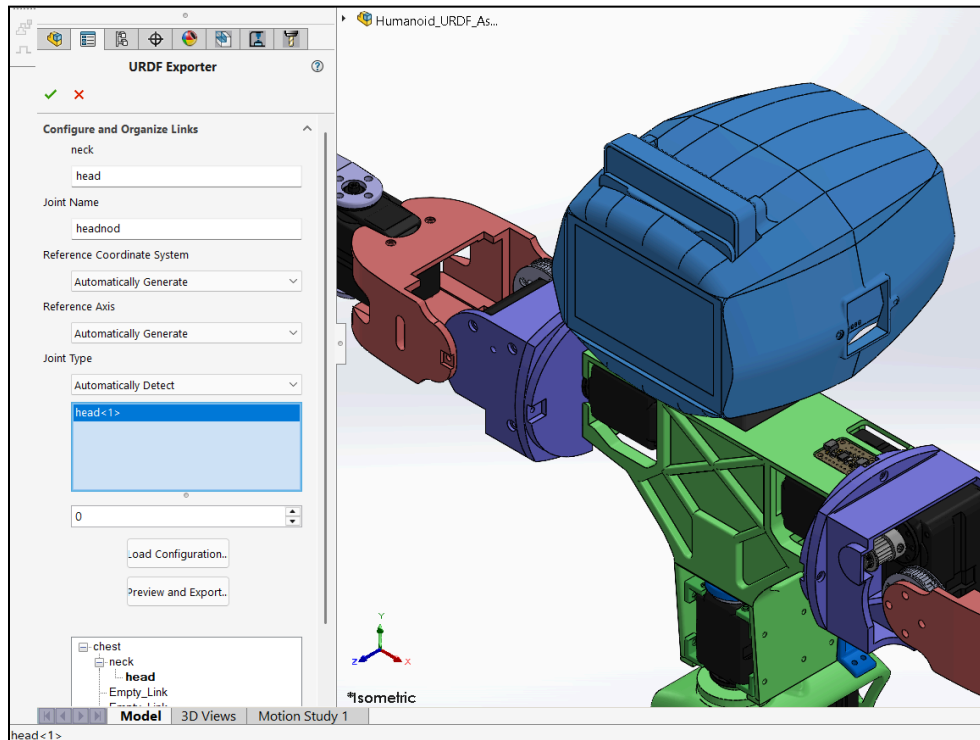
- Update the link and joint name in accordance with the naming conventions and select the link



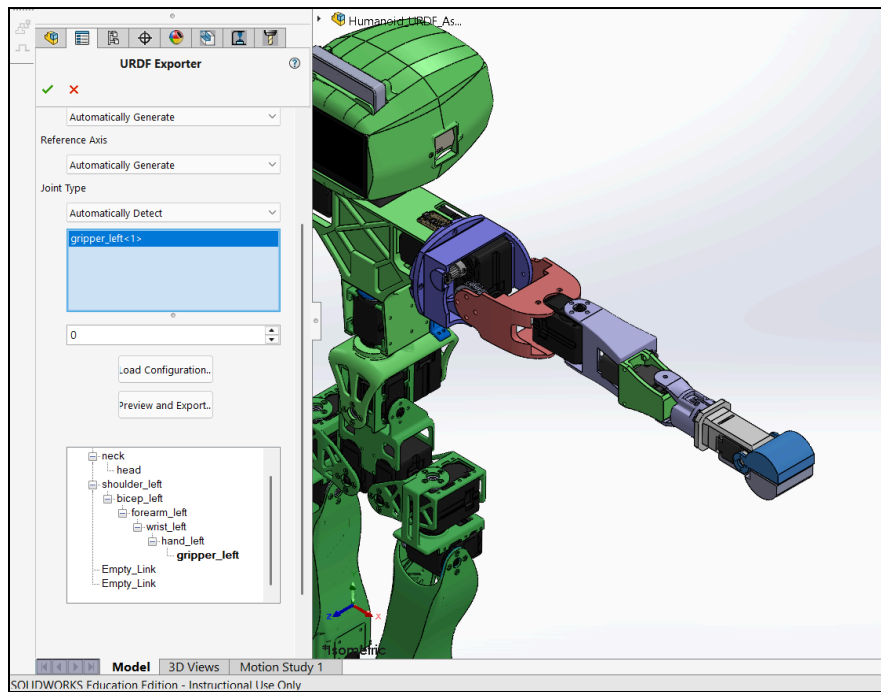
- Add a child link to the parent link



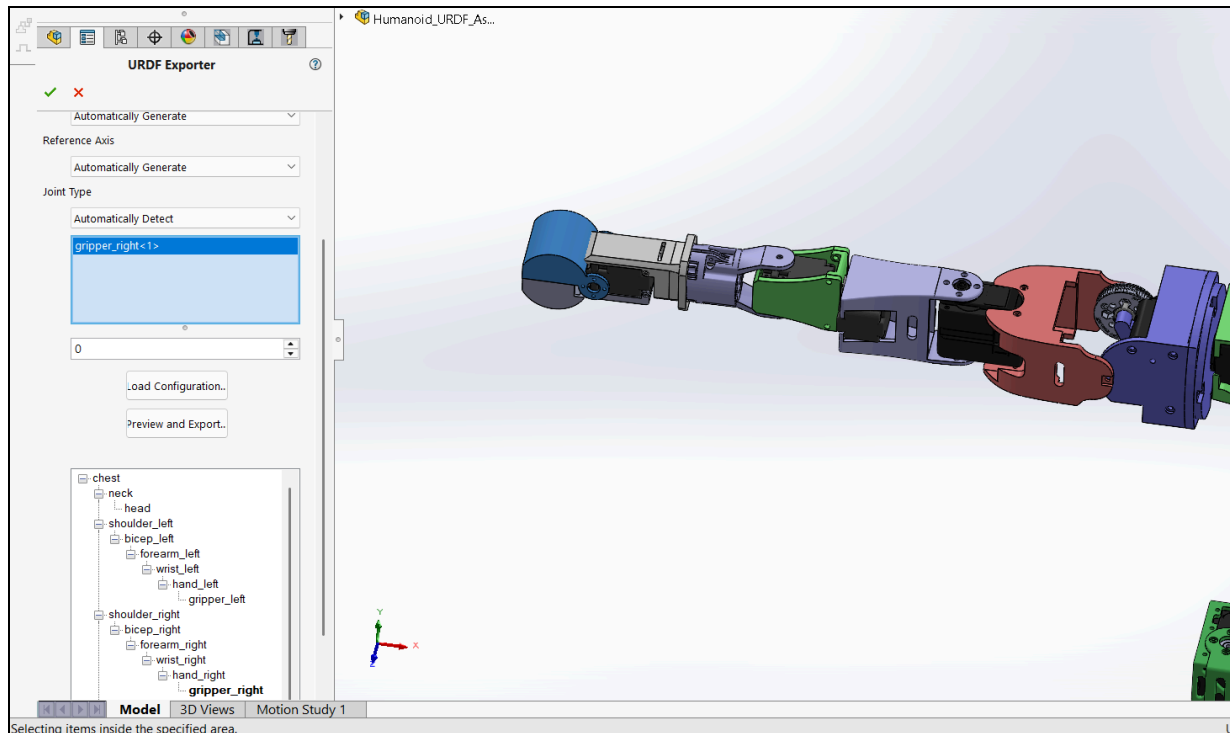
9. Select the child link, define the link name, define the joint name, and select the link model



10. Move on to the arms and the lower body, defining links in sequence according to naming conventions.

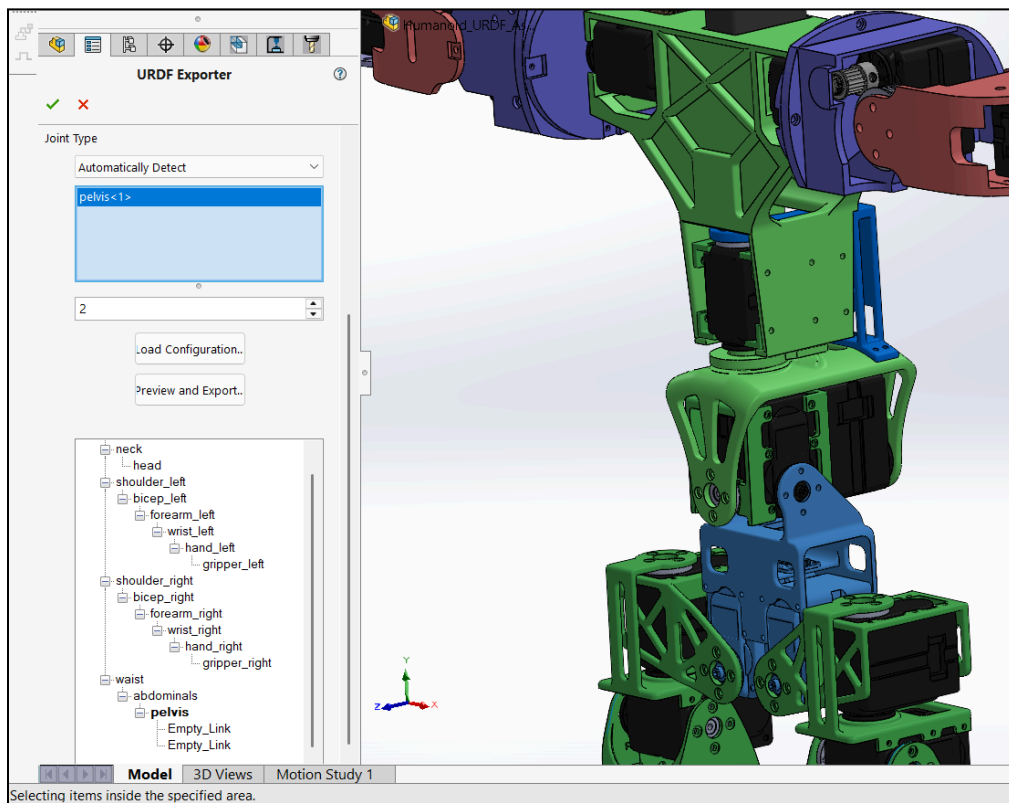


Left arm defined shoulder_left → gripper_left

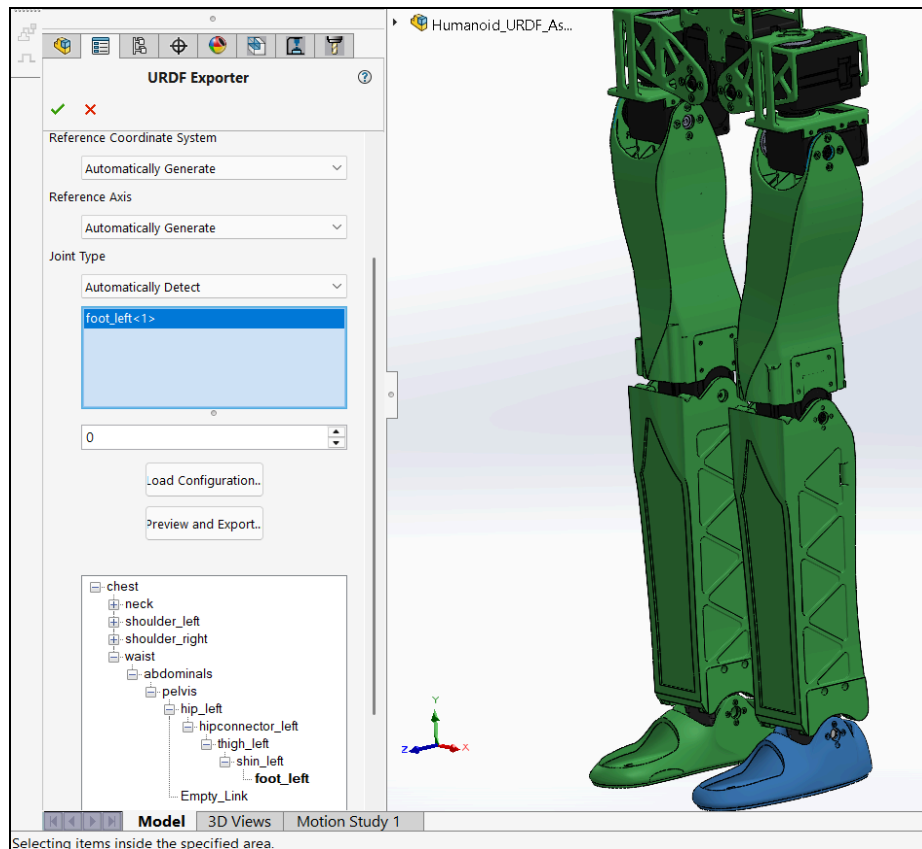


Right arm defined shoulder_right → gripper_right

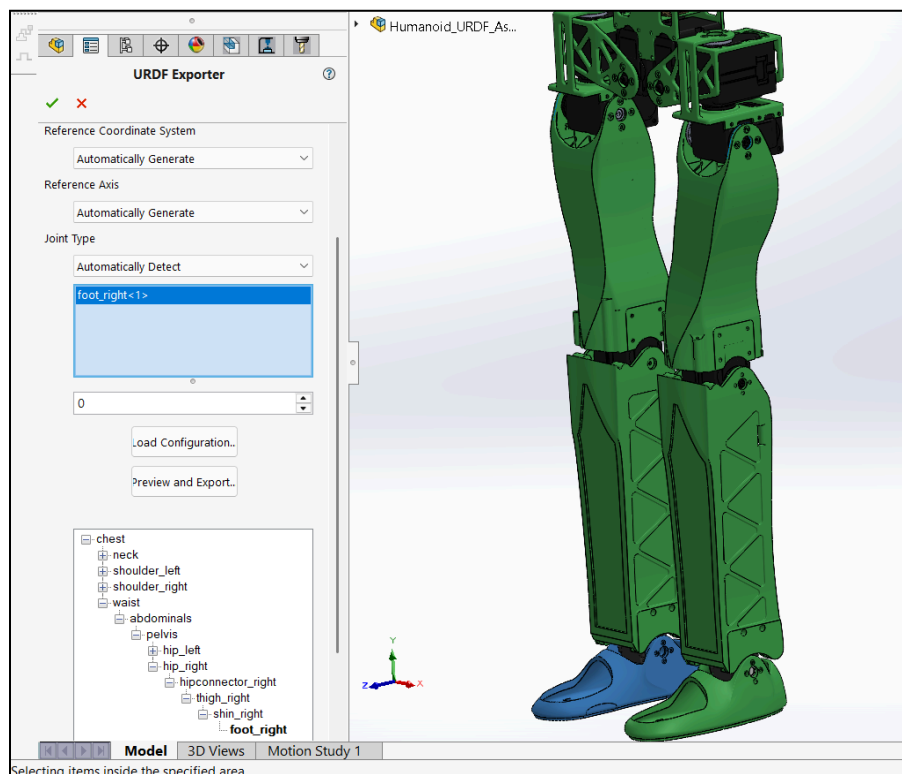
- When you get to a link with multiple child links, add the appropriate number of child links and continue.



Waist, abdominals, and pelvis defined. Pelvis has two child links

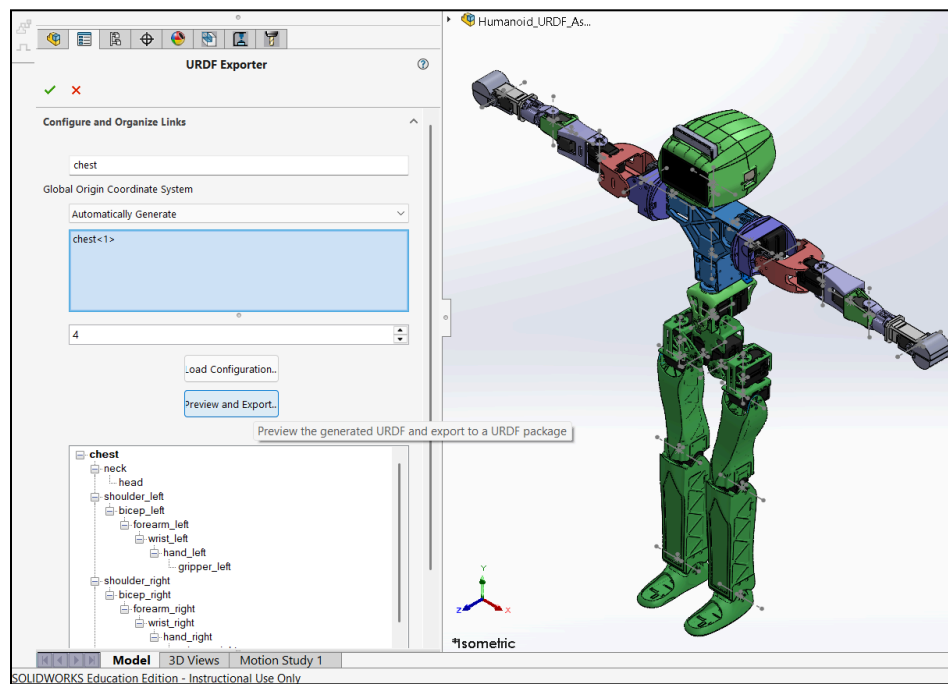


Left leg defined hip_left → foot_left

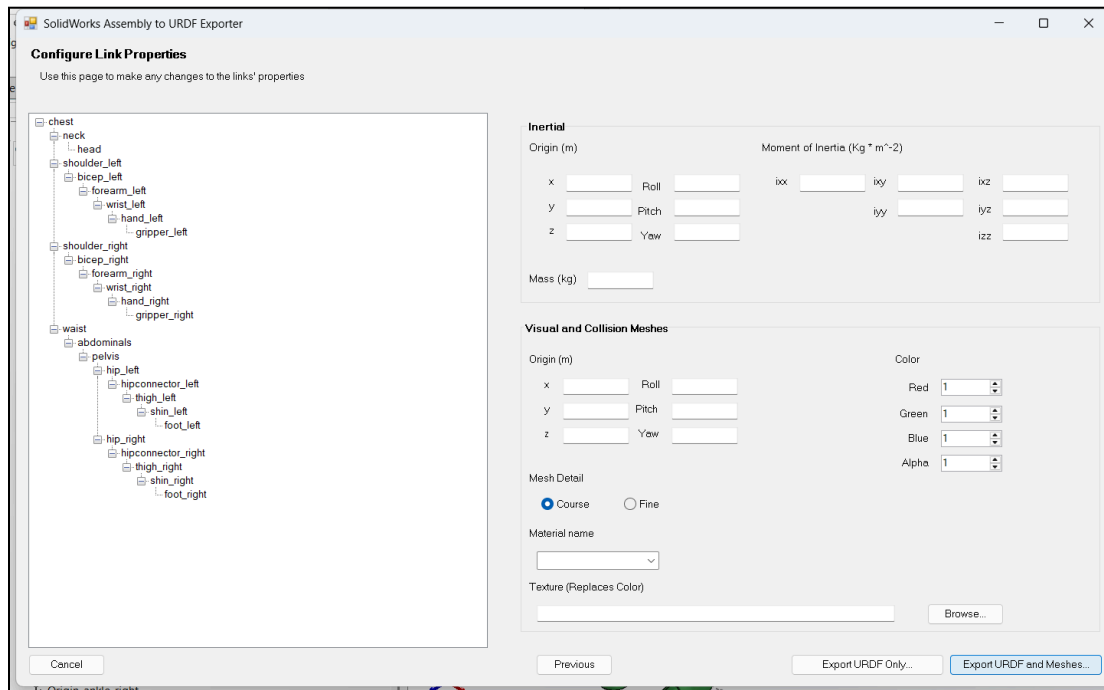


Right leg defined hip_right → foot_right

11. Once the entire chain is defined, click Preview and Export



12. Once the Axes and Origins have generated, fix any issues and click Export URDF and Meshes



13. Select location for export and click okay